

naive_bayes_iris

```
In [2]: from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
```

```
In [4]: X,y = load_iris(return_X_y=True)
X_train, X_test, y_train, y_test=train_test_split(X,y,test_size=0.5,random_state=0)
gnb=GaussianNB()
y_pred=gnb.fit(X_train, y_train).predict(X_test)
print(y_pred)
x_new=[[5,5,4,4]]
y_new=gnb.fit(X_train,y_train).predict(x_new)
print("predicted output for [[5,5,4,4]]:",y_new)
print("Naive bayes score:",gnb.score(X_test,y_test))
```

```
[2 1 0 2 0 2 0 1 1 1 1 1 1 1 1 0 1 1 0 0 2 1 0 0 2 0 0 1 1 0 2 1 0 2 2 1 0
 1 1 1 2 0 2 0 0 1 2 2 1 2 1 2 1 1 2 1 1 2 1 2 1 0 2 1 1 1 1 2 0 0 2 1 0 0
 1]
predicted output for [[5,5,4,4]]: [2]
Naive bayes score: 0.9466666666666667
```

```
In [ ]:
```